Name _____ Surname _____ Date ___/___/_____

Exam-style questions: solutions

Worked solutions. Answers are in italics.

Exercise 1

Rolling ball

A student times a ball rolling down a ramp five times. Her results are 1.84 s, 1.79 s, 2.31 s, 1.82 s and 1.80 s.

(a) Identify the anomalous result. [1]

2.31 s

(b) Calculate the mean time, ignoring the anomalous result. Give your answer to 3 significant figures. [2]

(1.84 + 1.79 + 1.82 + 1.80) ÷ 4 = 7.25 ÷ 4 = 1.8125

mean = 1.81 s

(c) Suggest one reason why her results are not all the same. [1]

Her reaction time in starting and stopping the stopwatch varies (or the ball is released slightly differently each time).

Exercise 2

The balance that isn't zero

With nothing on it, a digital balance reads 0.3 g. A student places a stone on the balance and it reads 25.6 g.

(a) State the name of this type of error. [1]

zero error (a systematic error)

(b) Calculate the correct mass of the stone. [1]

25.3 g

(c) The student weighs the stone five times and takes the mean. Explain whether this removes the error. [1]

No. Every reading is 0.3 g too high, so the same shift remains in the mean. Averaging reduces random errors, not systematic errors.

Exercise 3

A thermometer has a scale marked every 1 °C.

(a) State the smallest change in temperature that can be read reliably from this thermometer. [1]

0.5 °C

(b) The top of the liquid is halfway between the 23 °C and 24 °C marks. Record the temperature. [1]

23.5 °C

Exercise 4

Six students each time the same single run of a toy car over 1.00 m. Their results are: 2.46 s, 2.52 s, 2.49 s, 2.95 s, 2.43 s, 2.50 s.

(a) Identify the anomalous result. [1]

2.95 s

(b) Calculate the mean time, ignoring the anomalous result. [2]

(2.46 + 2.52 + 2.49 + 2.43 + 2.50) ÷ 5 = 12.40 ÷ 5 = 2.48

mean = 2.48 s

(c) Calculate the range of the results, ignoring the anomalous result. [1]

range = 2.52 − 2.43 = 0.09 s

(d) Suggest why the teacher asked all six students to time the same run, instead of six different runs. [1]

So that the differences between the results come only from the timing and not from differences in the run itself.

(e) Suggest one change to the method that would reduce the spread of the results. [1]

Use automatic timing (light gates), or a clear visual reference for starting and stopping the stopwatch.

Exercise 5

For each situation, state whether it causes a random error or a systematic error. Give a reason for each answer. [4]

(a) Each time, a student starts the stopwatch slightly early or slightly late.

Random error: sometimes the start is early and sometimes late, so the readings scatter above and below the true value.

(b) The end of a ruler is worn away, so its edge is at the 2 mm mark. A student always lines objects up with the edge.

Systematic error: every length is measured 2 mm too large, always in the same direction.

(c) A student always reads a measuring cylinder with her eye above the level of the liquid.

Systematic error: parallax always occurs in the same direction, so every reading is shifted the same way.

(d) Gusts of air sometimes speed up and sometimes slow down a falling paper cone.

Random error: the gusts sometimes speed up and sometimes slow down the cone, unpredictably.

Exercise 6

The true time of a swing is 2.50 s. A student's reactions are not perfect.

(a) He starts the stopwatch 0.20 s late and stops it exactly on time. State whether his reading is too long or too short, and by how much. [1]

too short, by 0.20 s (it reads 2.30 s)

(b) Next time he starts it 0.20 s late and stops it 0.20 s late. What does the stopwatch read? [1]

2.50 s

(c) Then he starts it 0.20 s late and stops it 0.30 s late. Calculate the reading. [1]

2.50 − 0.20 + 0.30 = 2.60 s

(d) Explain why reaction times that change from one reading to the next produce a random error. [1]

The error changes unpredictably in size and direction from one reading to the next, so the readings scatter above and below the true value.